IRREGULARITIES – CHIEF (Coursework only)

If you suspect an assessment irregularity (e.g. plagiarism, collusion or multiple submission) please list the student ID and potential issue, otherwise leave blank.

STUDENT ID(s)	SUSPECTED ISSUE
	None

NUMERICAL ANSWERS – CHIEF (Exam only, EXCEPT RESITS, published to students)

Q1

(a) None
(b) None
(c) 13.21 kN/m
The boom axial forces and the shear flow clockwise of the boom are given below.

boom, i	ΔP_i	$\sum_{j=1}^{i} \Delta P_j$	shear flow (clockwise of boom)	q_i (N/m)
1	2070	2070	$q_{16} - 2070$	5102
2	2847	4917	$q_{16} - 4917$	2255
3	3190	8108	$q_{16} - 8108$	-936
4	3048	11155	$q_{16} - 11155$	-3983
5	2441	13597	$q_{16} - 13597$	-6425
6	1463	15060	$q_{16} - 15060$	-7888
7	262	15323	$q_{16} - 15323$	-8150
8	-978	14344	$q_{16} - 14344$	-7172
9	-2070	12274	$q_{16} - 12274$	-5102
10	-2847	9427	$q_{16} - 9427$	-2255
11	-3190	6237	$q_{16} - 6237$	936
12	-3048	3189	$q_{16} - 3189$	3983
13	-2441	747	$q_{16} - 747$	6425
14	-1463	-716	$q_{16} + 716$	7888
15	-262	-978	$q_{16} + 978$	8150
16	978	0	$q_{16} - 0$	7172

(d)
(e) 0.274mm

Q2

(a) None
(b)

boom	y (mm)	σ_{xx} (MPa)
1,2	152	1.404
3	102	0.942
4,5	-152	-1.404
6	-102	-0.942

(c)

panel	shear flow, q (N/m)	thickness, t (mm)	shear stress (MPa)
12	2013	0.92	2.19
23	0	0.56	0
14 (inner)	1146	2.03	0.56
14 (outer)	867	0.56	1.55
25	2013	1.63	1.23
45	2013	0.92	2.19
56	0	0.56	0

Twist, $\theta = 2.80 \times 10^{-4}$ radians/m
Torsion constant, $J = 1.590 \times 10^{-4}$ m^4
(d) None

ASSESSMENT CHECKLIST 2

MARKS FINALISATION – CHIEF AND PROOF

We confirm that:

- The marking and checking of the above assessment have been completed in accordance with the Manual of Assessment Procedures.
- The marks conform to the advertised marks matrix, Senate Marking Scale and appropriate UG/MScT Level descriptors.
- Marks have been entered onto the spreadsheet by the Chief, moderated by the Proof and checked by the Checker.

Chief: Will Coombs **Date:** 2nd June 2020

Proof: ___Charles Augarde_____ **Date:** ___02 June 2020__

Assessment Average:

Are marks within the expected bands for this level? **Yes**
(NOT required for resit examinations and examinations with less than five submissions)

Chief's Report (Exams and Coursework, published to students and external examiners – NOT required for resit examinations)

- Brief discussion (if necessary) as to why marks are outside expected range.
- Cohort feedback: What went well? What were the common errors or misunderstandings?

The average mark is within the bands.

Feedback

Q1: Good performance with an average of 68%. Part (a) was reasonable but often students were not very detailed in their answers, just giving a very brief summary of the notes. Part (b) was answered well but many students were not clear on the assumptions that they were making, such as about the value of the local second moment of area. Part (c) was straightforward for most students but some failed to give a clear statement about the direction of the shear flow. Part (d) was answered very well by most students, however some were not clear about units or about the locations of the various shear flows through the structure. Part (e), straightforward for most students by combining the results from (c) and (d).

Q2: Reasonable performance with an average mark of 63%. In general, there was a lack of explanation of the assumptions made and students lost marks due to this. Parts (a), (b) and (c) were answered well with a lot of the students getting the correct answers or going through the correct working. In Part (c) some students only gave the shear flow, not the shear stress as asked for in the question. Part (d) was answered poorly and students failed to acknowledge the differences between a multi-celled and single-celled structure when calculating the warped shape.

Chair BoE: ____Claudio Balocco_____ **Date**___12/6/2020____